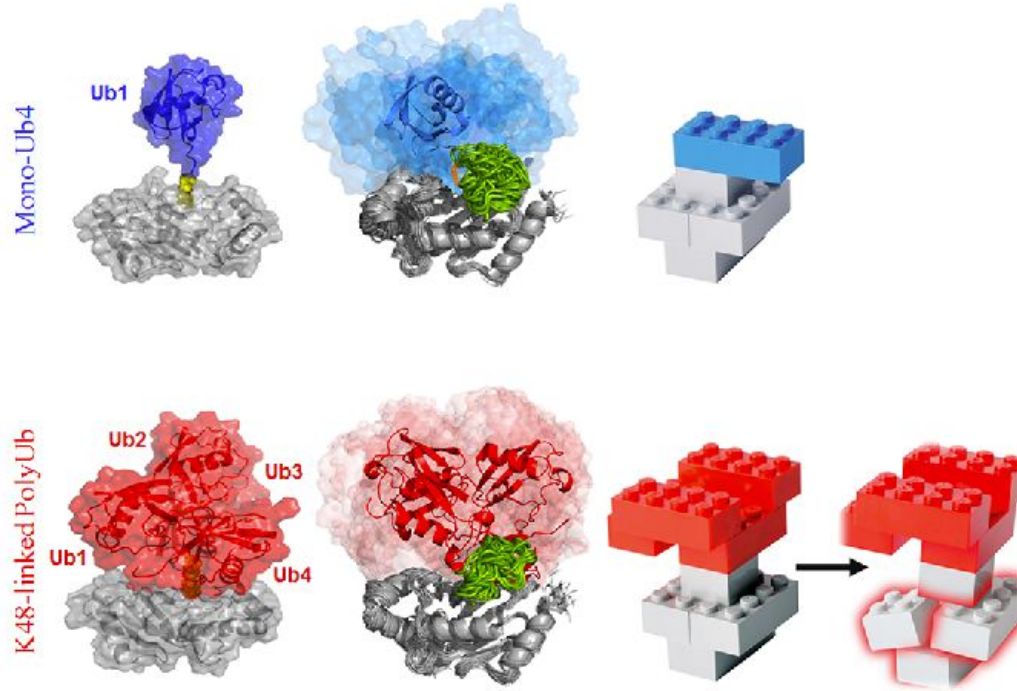


AlphaFold 2:

*How domain-science-informed neural architectures
can matter more than big data, through the lens of protein folding*

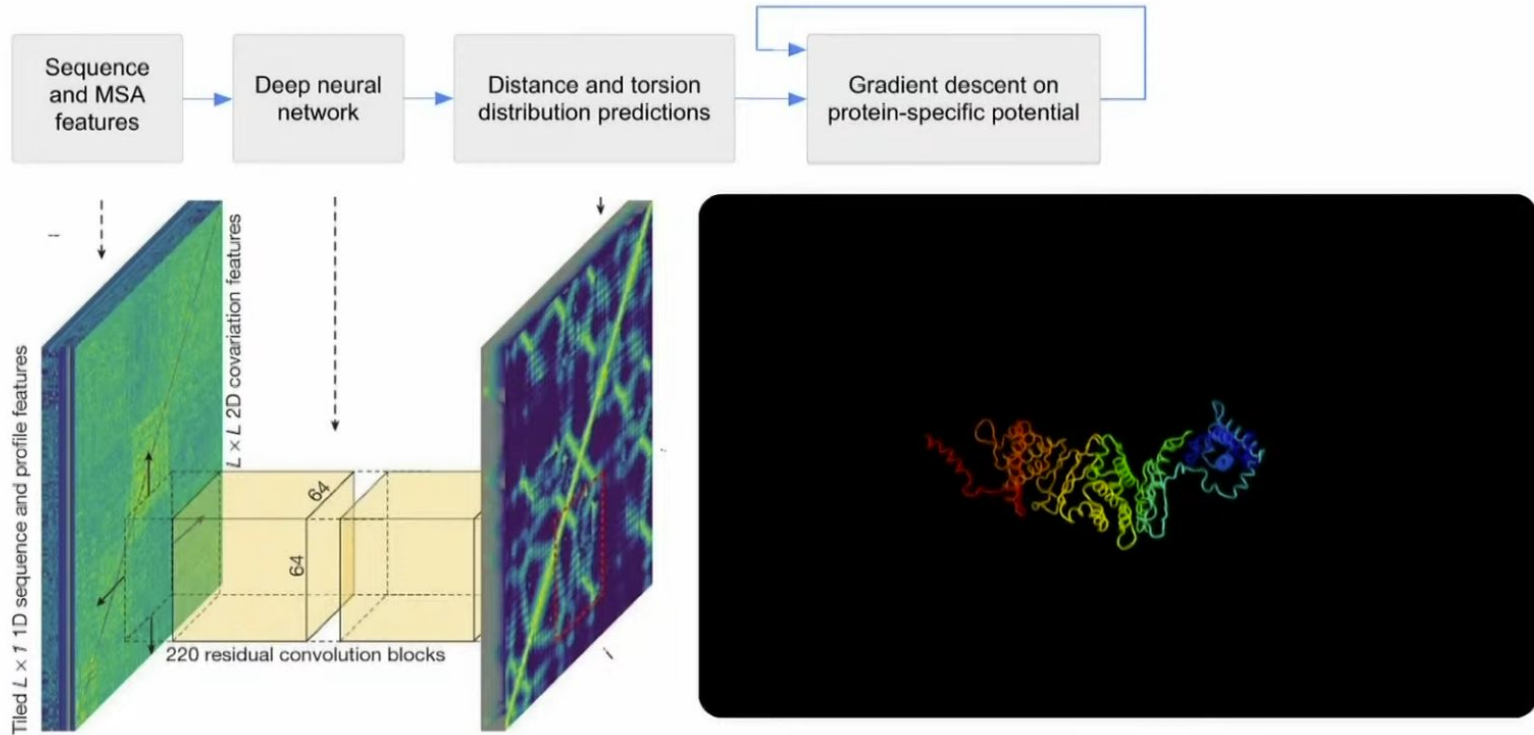
Introduction

Proteins Are Bio-LEGO-Like Structures



“Protein Assembly and Building Blocks: Beyond the Limits of the **LEGO** Brick **Metaphor**.” by Y. Levy

AlphaFold 1 (2018)



AlphaFold 1 was more like a **vision network** applied to distogram maps.

AlphaFold 1 vs AlphaFold 2 Performance

Median Free-Modelling Accuracy

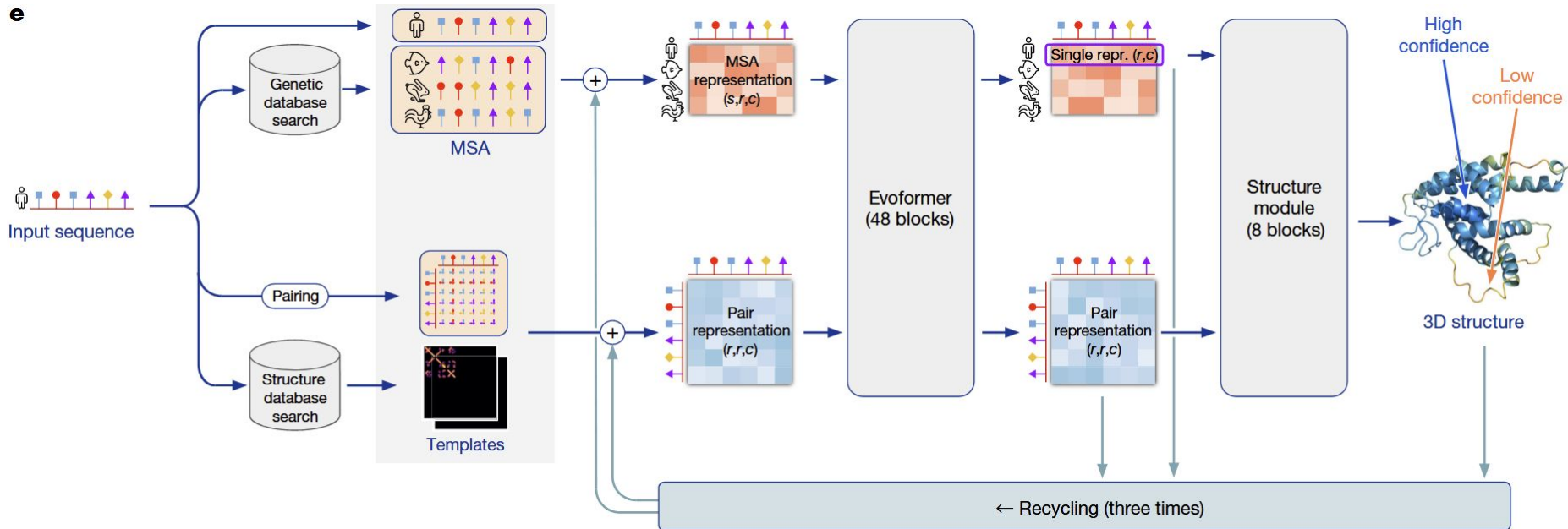


CASP is a **biennial international competition** that evaluates how well computational methods can **predict the 3D structure of proteins** from their amino-acid sequence.

AlphaFold 2

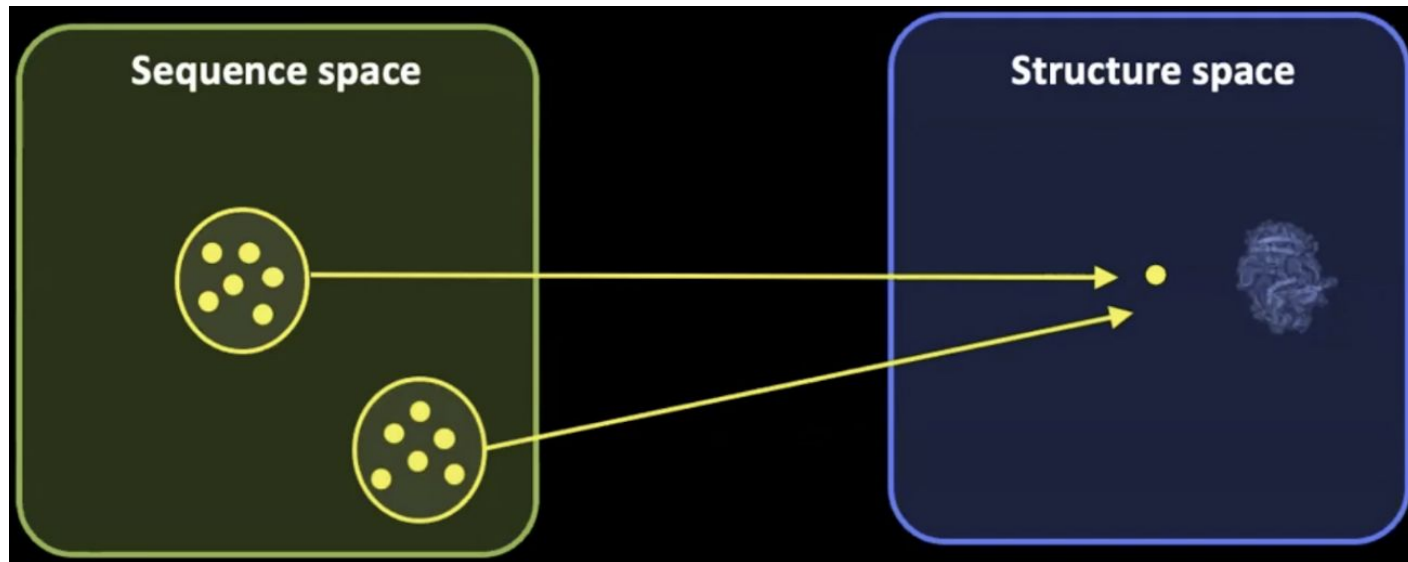
AlphaFold 2 (2020)

e



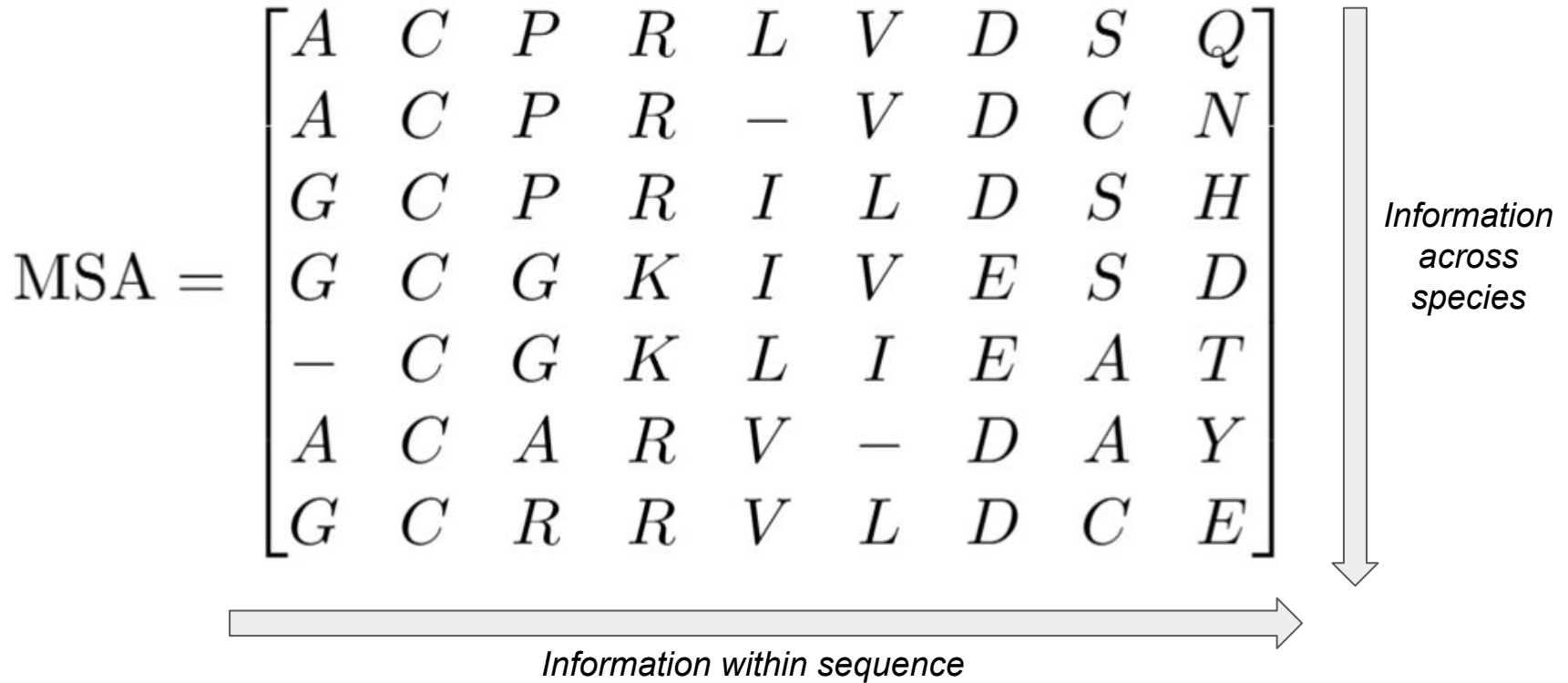
“Highly accurate protein structure prediction with AlphaFold”, by John Jumper et al.

Sequence-Structure Mapping

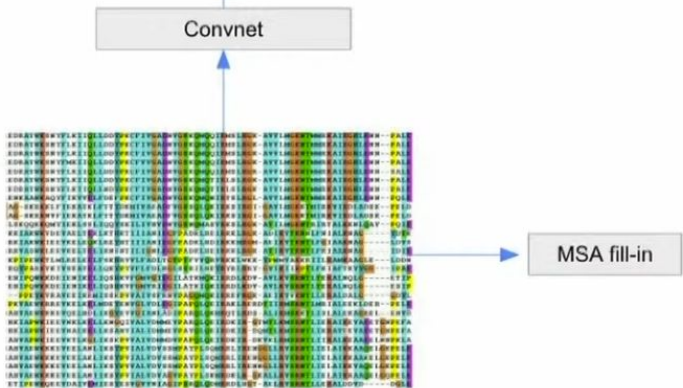
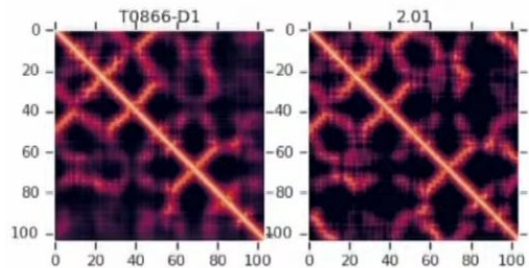


- Many different amino-acid sequences can converge to a similar 3D fold, so **sequence space is much larger** than **structure space**.
- This **many-to-one mapping** can actually **aid generalization**: diverse sequences may share the same overall fold, although predicting fine structural details remains challenging.

Multiple Sequence Alignments (MSA)

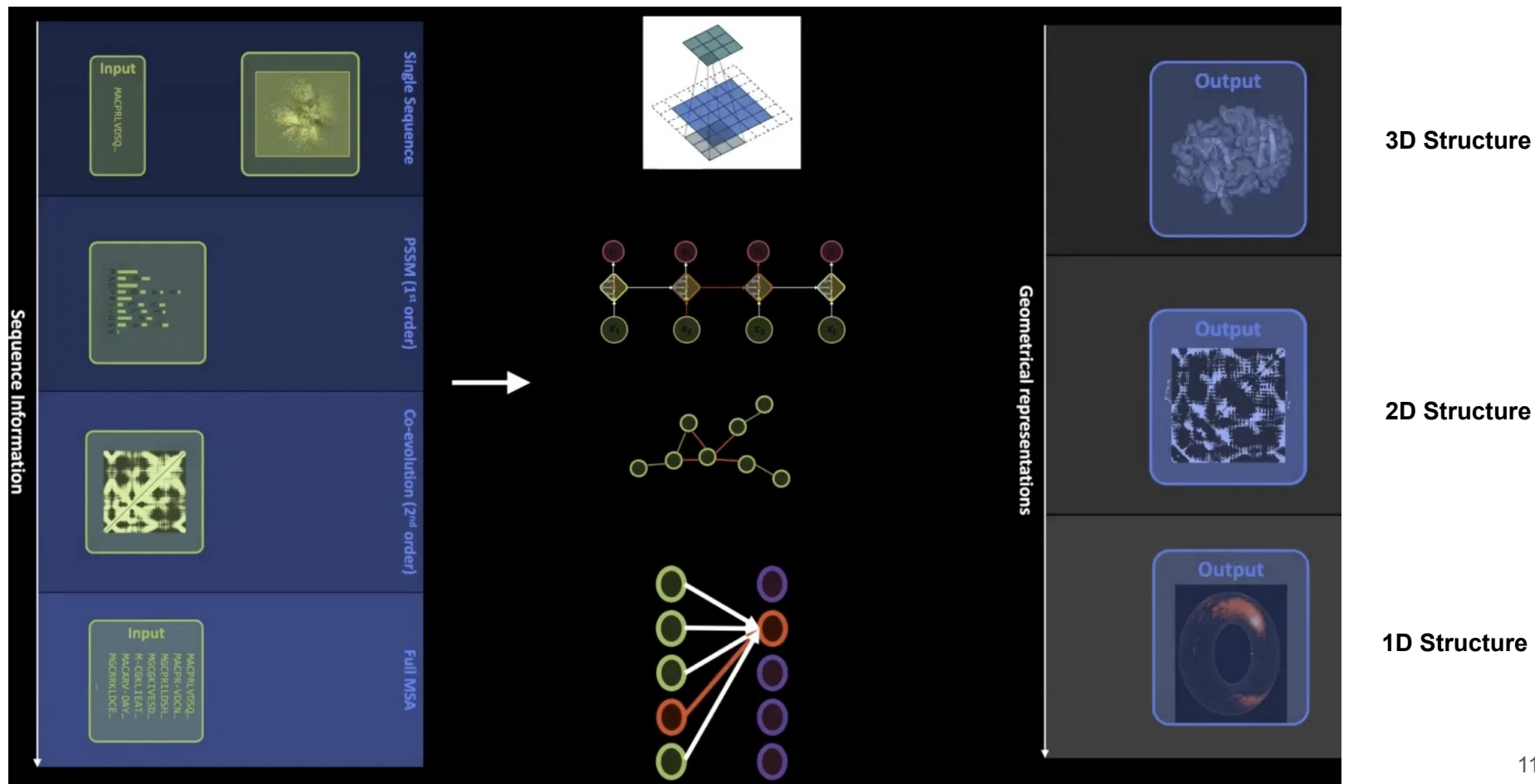


Using Evolutionary Data



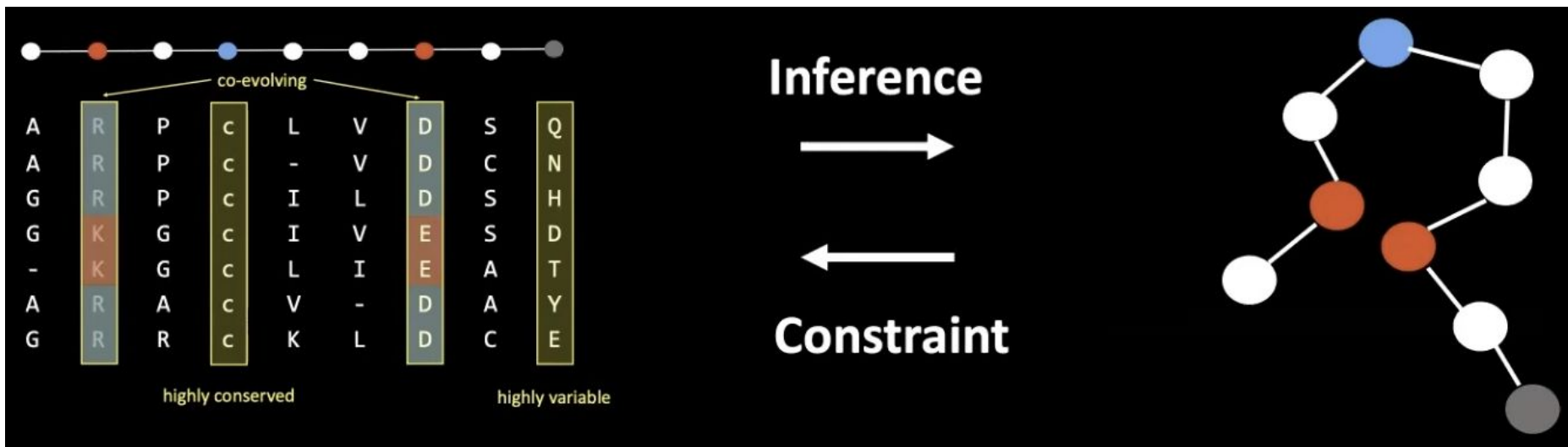
- BERT-style models on MSAs learned to **recover masked evolutionary information**
- **Axial attention alternates between processing within sequences and across sequences**
- This architectural choice **aligned the network operations with the biological structure of the problem**

AlphaFold 2 Links 1D, 2D, and 3D Representations



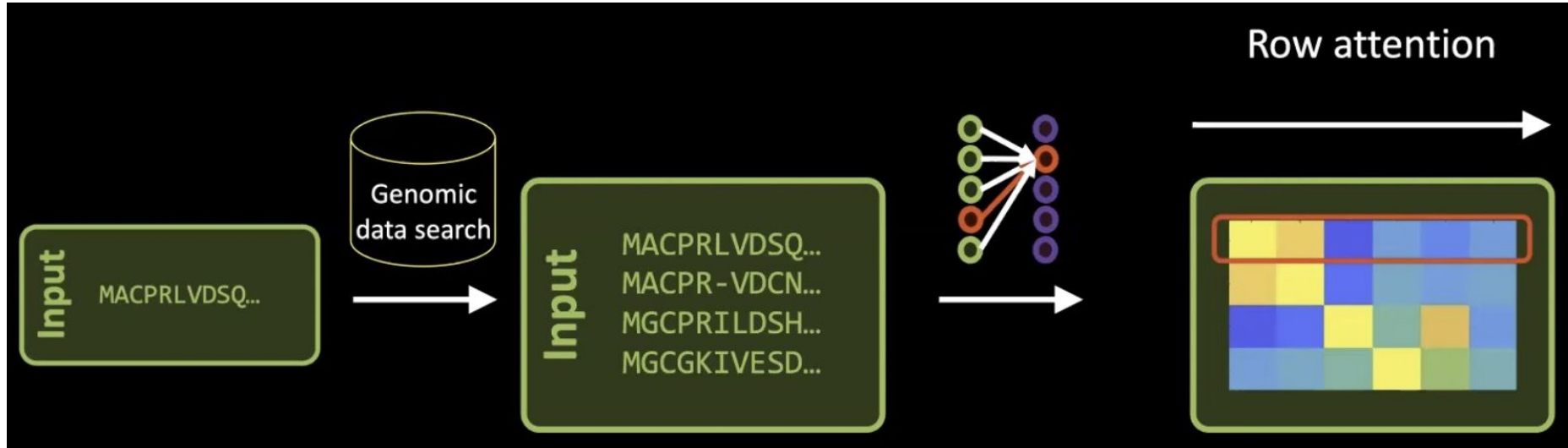
Modelling Co-Evolution

The **key inductive bias** we want to model is the idea of **co-evolution**

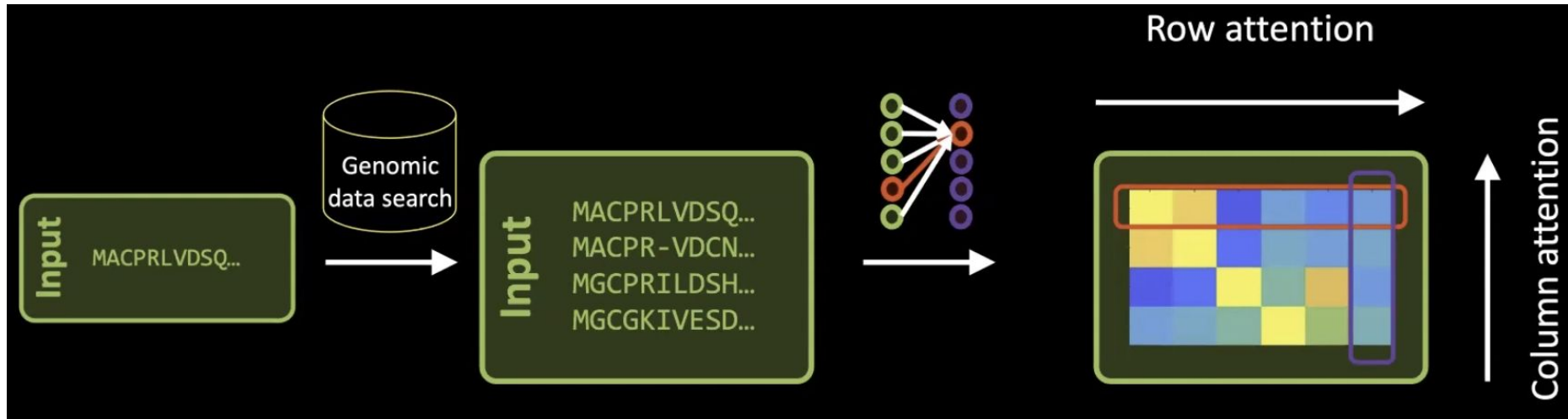


Note that the **red** tokens are the **residues**
(A residue is an amino acid unit within a protein chain)

Row Attention Integrates Information Within a Sequence



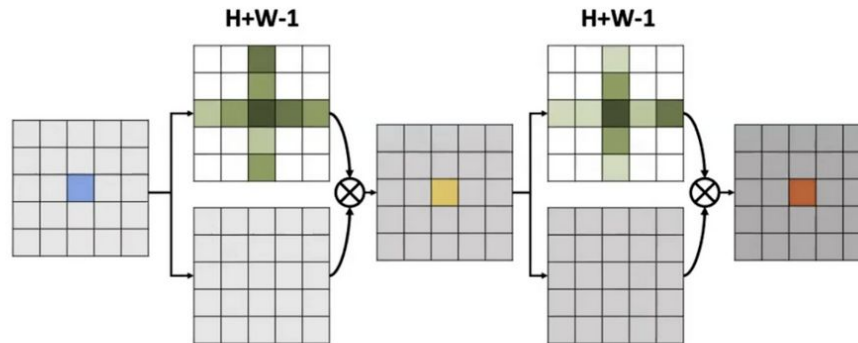
Column Attention Integrates Information Across Sequences



Axial Attention Alternates Between Rows and Columns



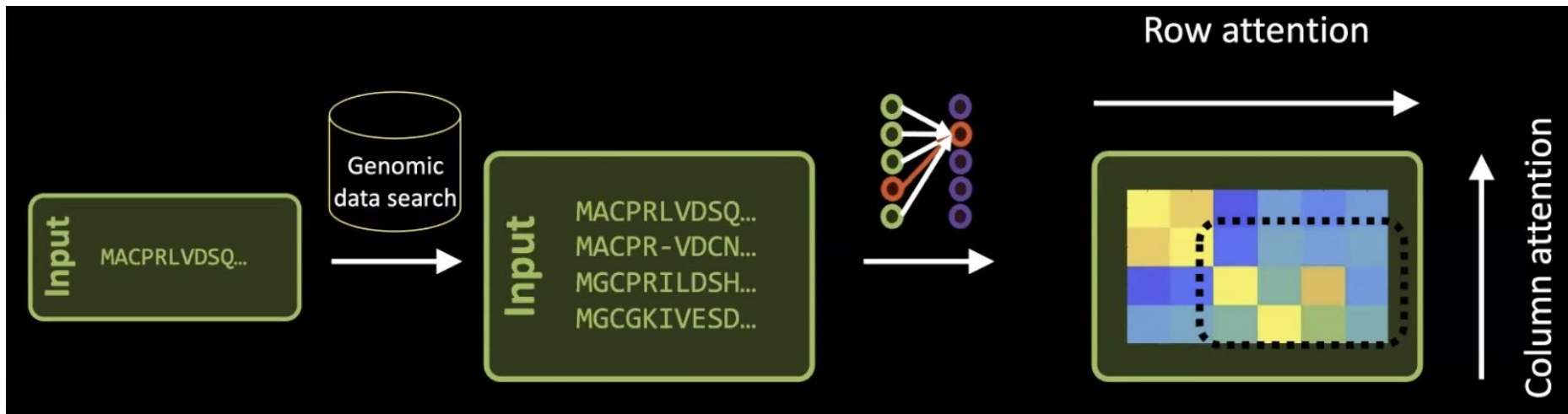
***Criss-cross** attends to the row and column of each position **simultaneously**. **Axial** attention **alternates**: first all **rows**, then all **columns**. Both achieve global coverage, but axial is simpler.*



(b) Criss-Cross Attention block

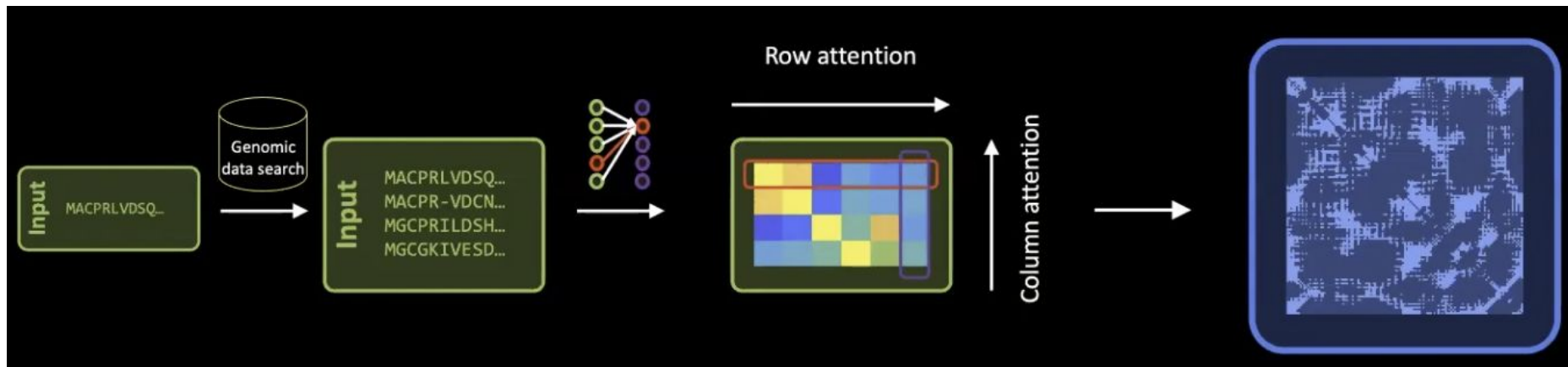
- AlphaFold 2 uses **axial attention** by **alternating row** and **column** attention
- This **separates processing** across sequences from processing along residue positions

Cropping Makes Training Tractable



AlphaFold uses cropping **during training**, but **not at inference** time.

The Evoformer First Predicts Pairwise Distances



First conceptual step is the prediction of a distogram. The **distogram may indicate the geometry** or fold of the protein.

*Note → A **distogram** is a matrix where entry (i, j) represents the predicted distance between residue i and residue j , specifically between their C^α atoms (the central carbon of each residue's backbone). So for a protein with L residues, you get an $L \times L$ matrix.*

MSA Transformer Helped Establish the Design Direction



bioRxiv
THE PREPRINT SERVER FOR BIOLOGY

MSA Transformer

Roshan Rao, Jason Liu, Robert Verkuil, Joshua Meier, John F. Canny, Pieter Abbeel, Tom Sercu, Alexander Rives

doi: <https://doi.org/10.1101/2021.02.12.430858>

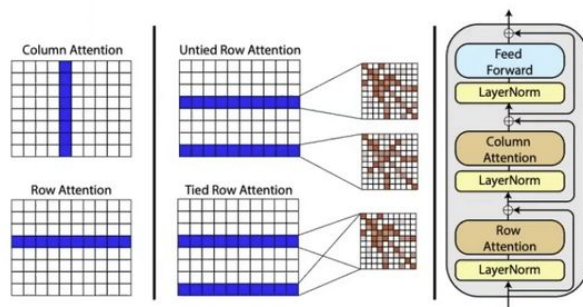
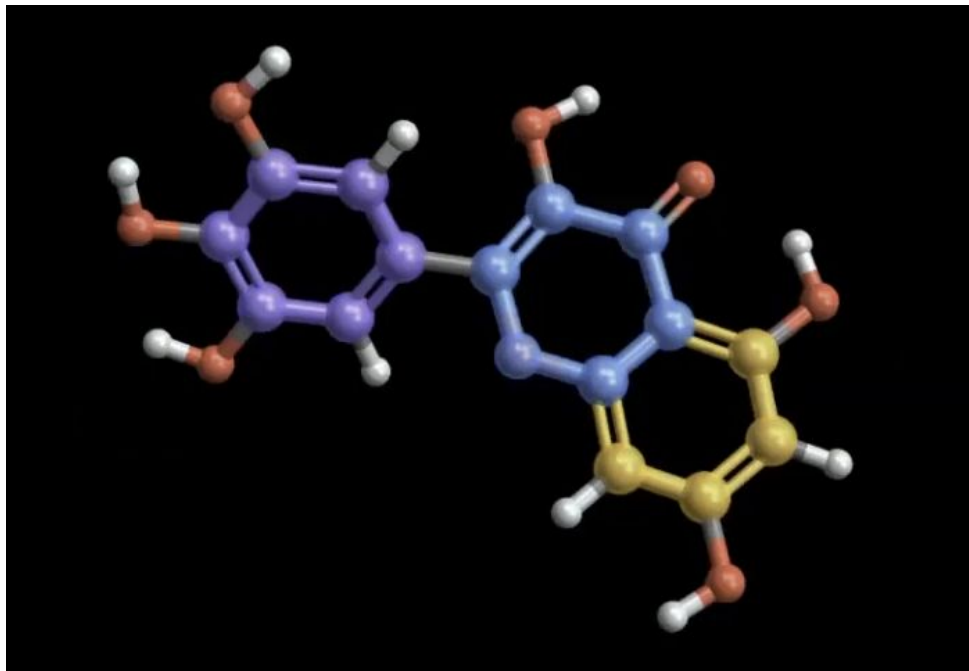


Table 3. Supervised contact prediction on CASP13 and CAMEO (long-range precision). *Uses outer-concatenation of the query sequence representation as features. †Additionally uses the row attention maps as features.

Model	CASP13-FM		CAMEO	
	L	L/5	L	L/5
trRosetta _{base}	45.7	69.6	50.9	75.5
trRosetta _{full}	51.8	80.1	53.2	77.5
Co-evolutionary	40.1	65.2	47.3	72.1
ProTrans-T5	25.0	41.4	40.8	63.3
ESM-1b	28.2	50.2	44.4	68.4
MSA Transformer*	54.5	80.2	53.6	78.0
MSA Transformer†	54.6	77.5	55.8	79.1

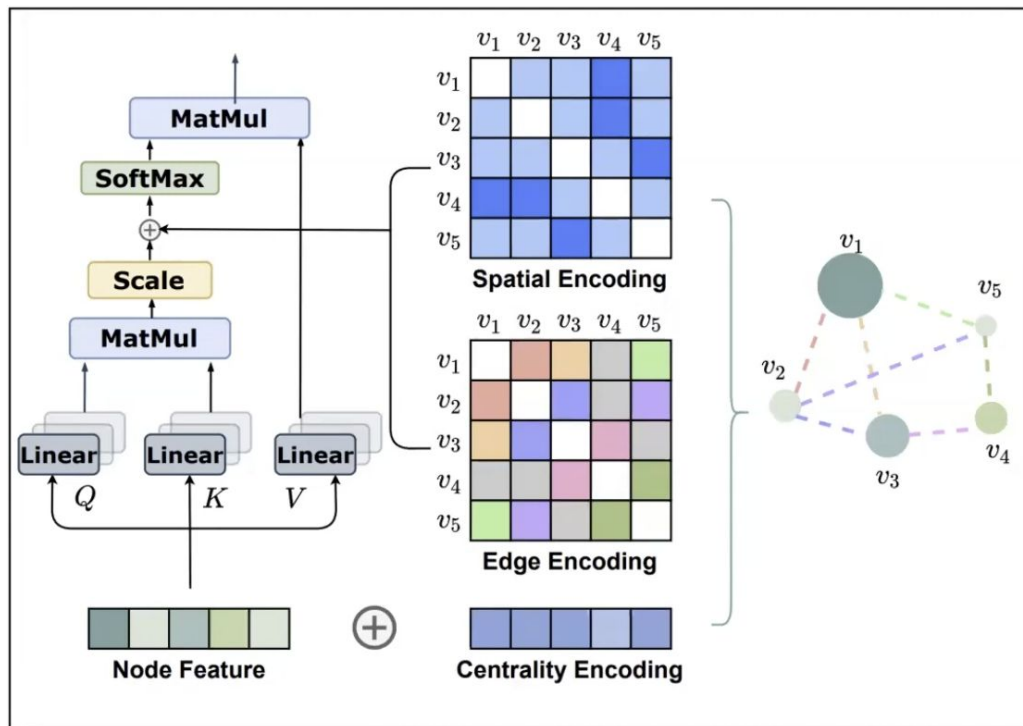
Proteins “are” Graphs

A chemical molecule can be represented as a graph, so we need to **encode relational structure**



Standard Transformers do not model edge information explicitly

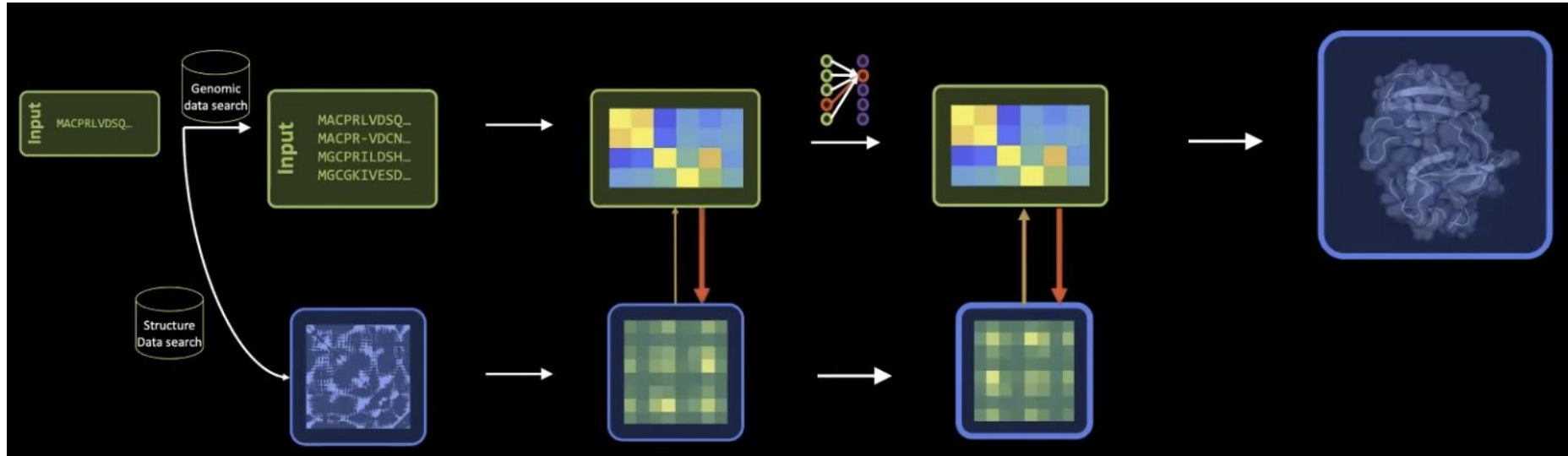
Transformer Graph Representations



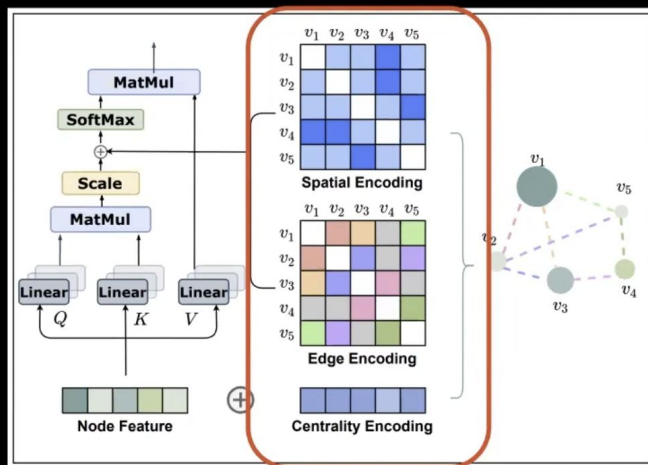
Extending Transformers to encode Graph Edges

“Do Transformers Really Perform Bad for Graph Representation?” by C. Ying et al., NeurIPS 2021

MSA and Pair Representations Exchange Information



Attention Bias Can Encode Pairwise Edge Information

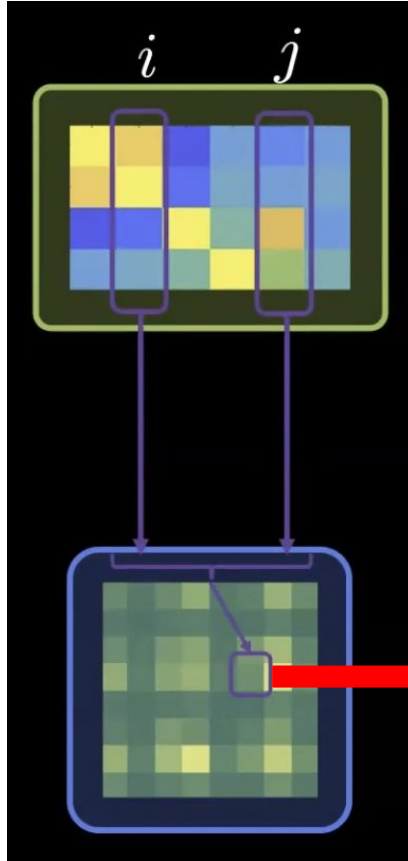


$$a_{ij} = \text{softmax}(q_i^T k_j + b_{ij})$$

Bias b == Edge Information

C. Ying et al., Do Transformers Really Perform Badly for Graph Representation? NeurIPS 2021

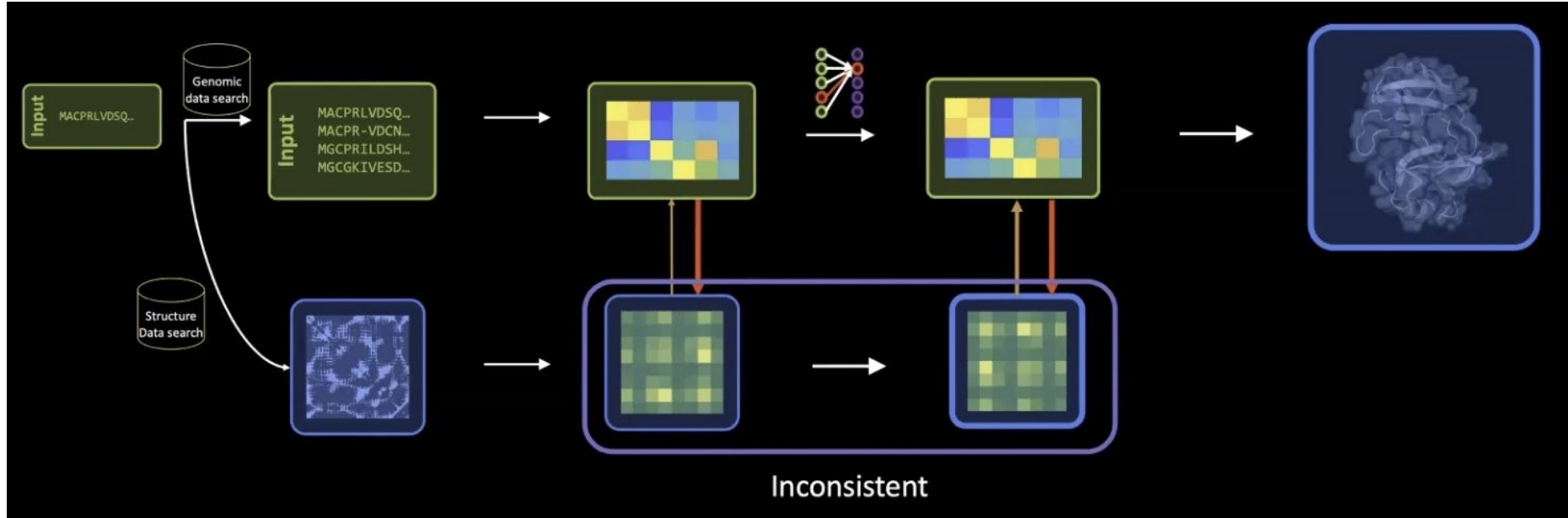
The Pair Representation Is Updated from the MSA



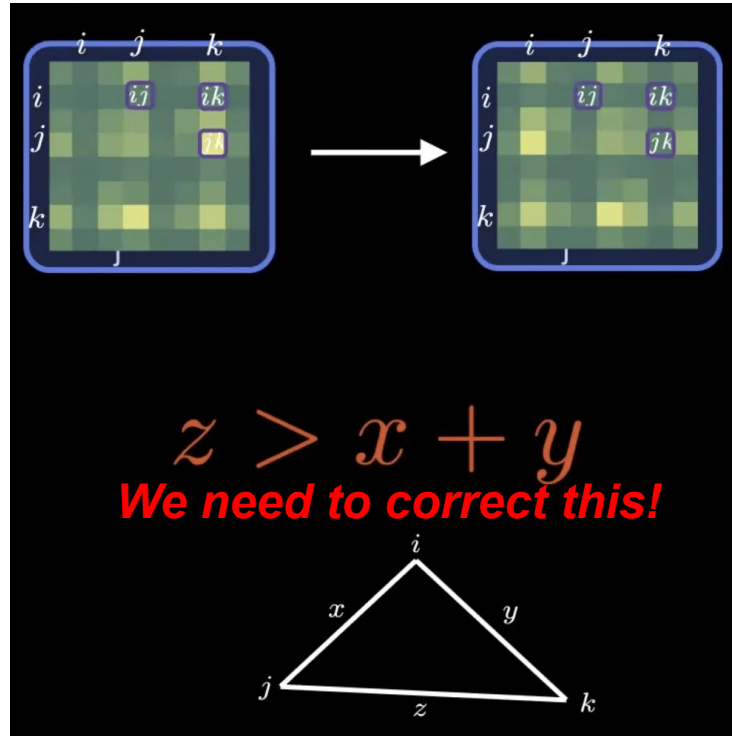
$$o_{ij} = \text{mean} (a_i \otimes b_j)$$

Take the **mean** of the **outer product** of 2 columns of the MSA.

Pairwise Predictions Must Be Geometrically Consistent



Triangle Constraints Enforce Geometric Consistency

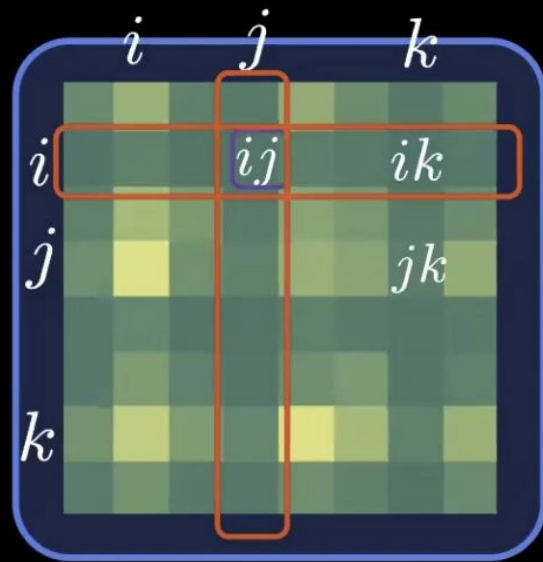


We must **obey the triangle inequality** $\rightarrow z \leq x + y$

AlphaFold 2 Enforces Triangle Constraints Through Smooth Updates

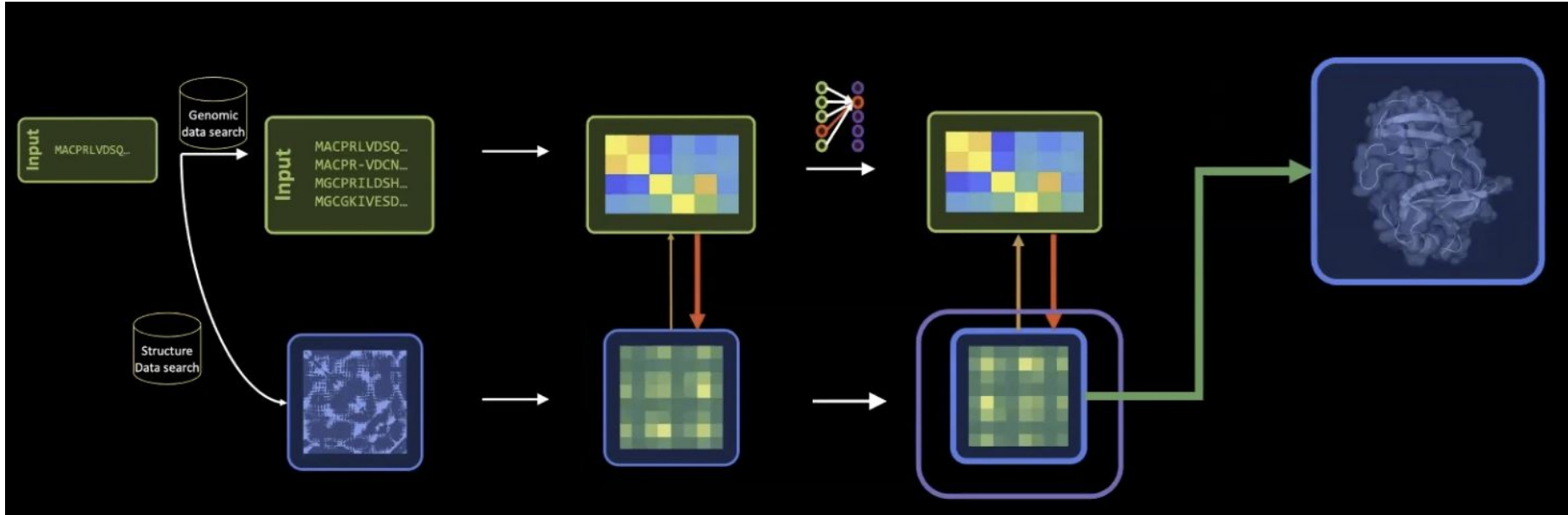
Expensive!
But worth it.

$$z_{ij} \leftarrow f\left(\sum_k a_{ik} b_{jk}\right)$$



Triangle consistency is encouraged through **smooth** updates of the 2D pair representation

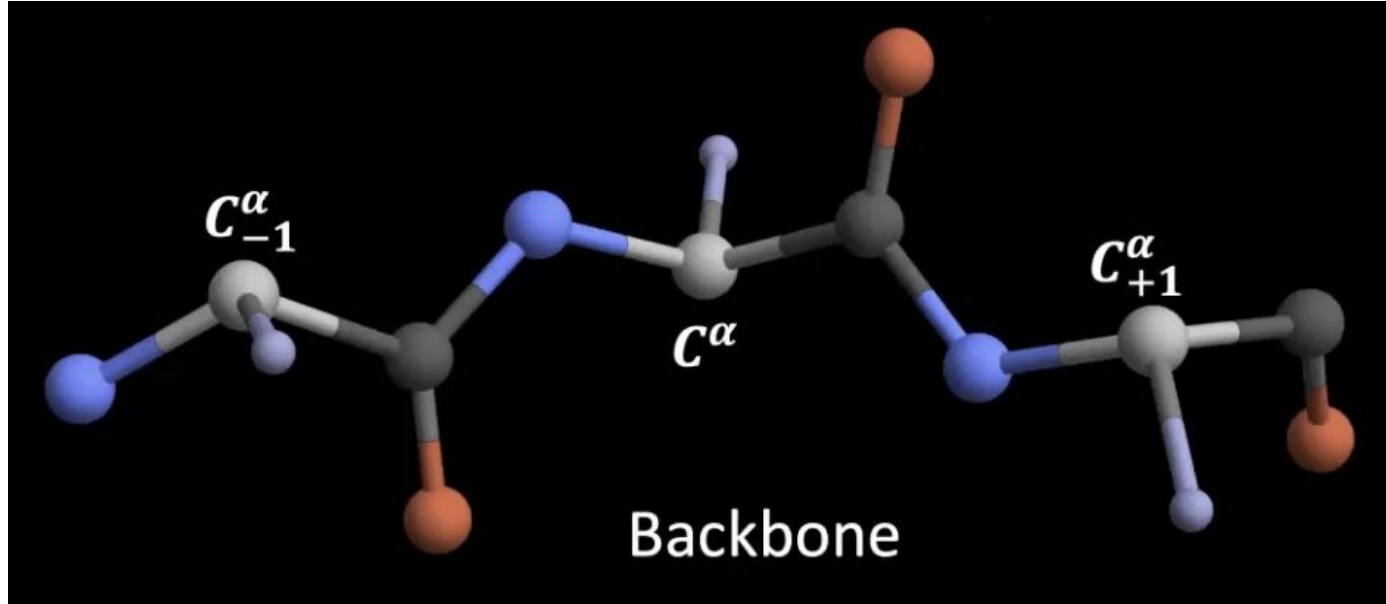
The Structure Module Converts Representations into 3D Coordinates



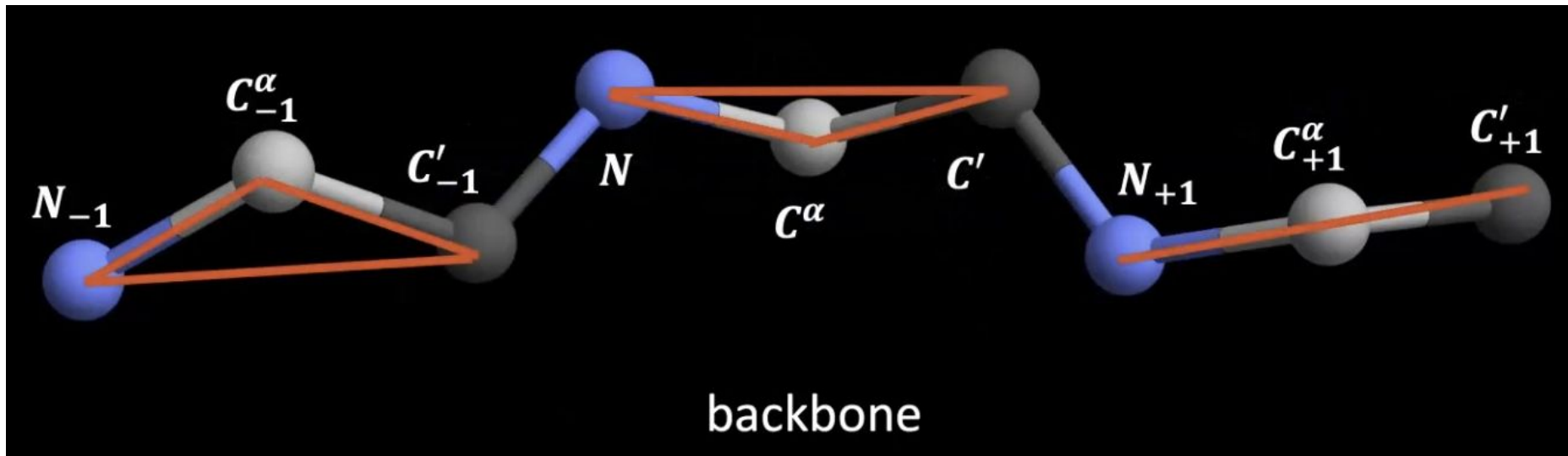
To produce 3D structure, AlphaFold uses the **Pair Representation** together with **the First Row of the MSA**, corresponding to the target sequence.

There are **many more pairwise relationships than there are coordinates to find**, so the structure is heavily constrained → i.e. the model has more than enough information to pin down the 3D shape.

The Protein Backbone Provides the Structural Scaffold

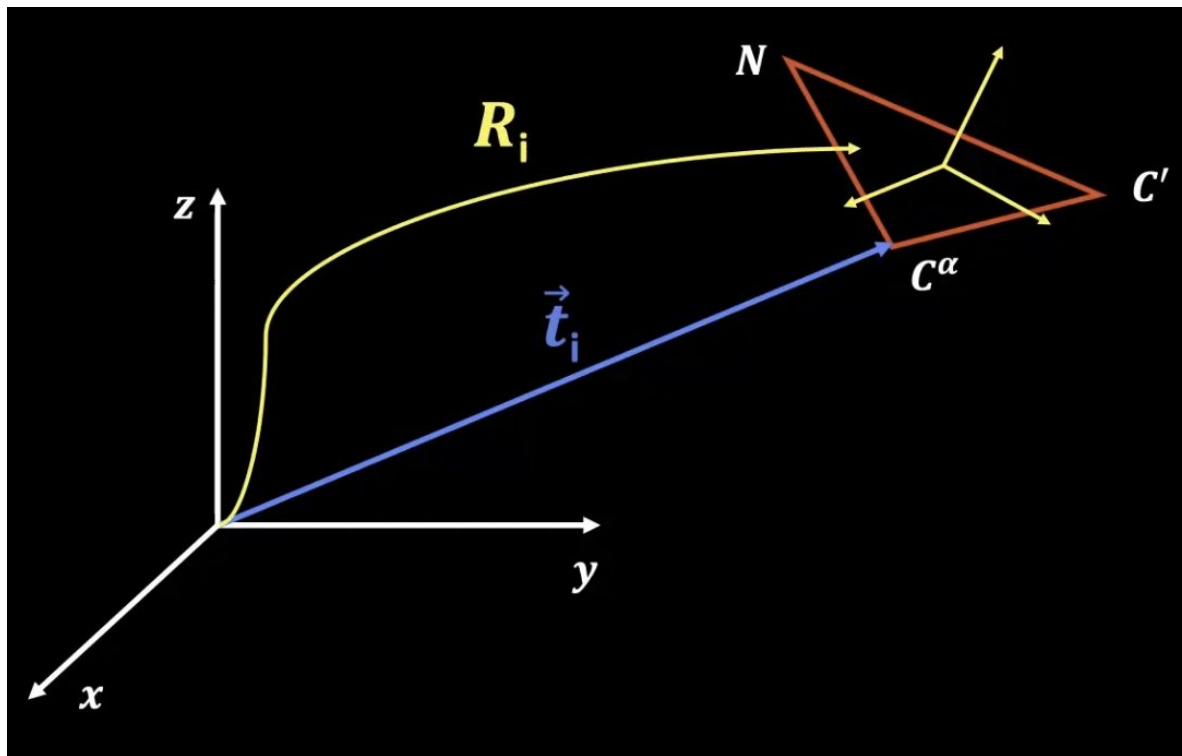


Each Residue Is Represented as a Rigid Local Frame



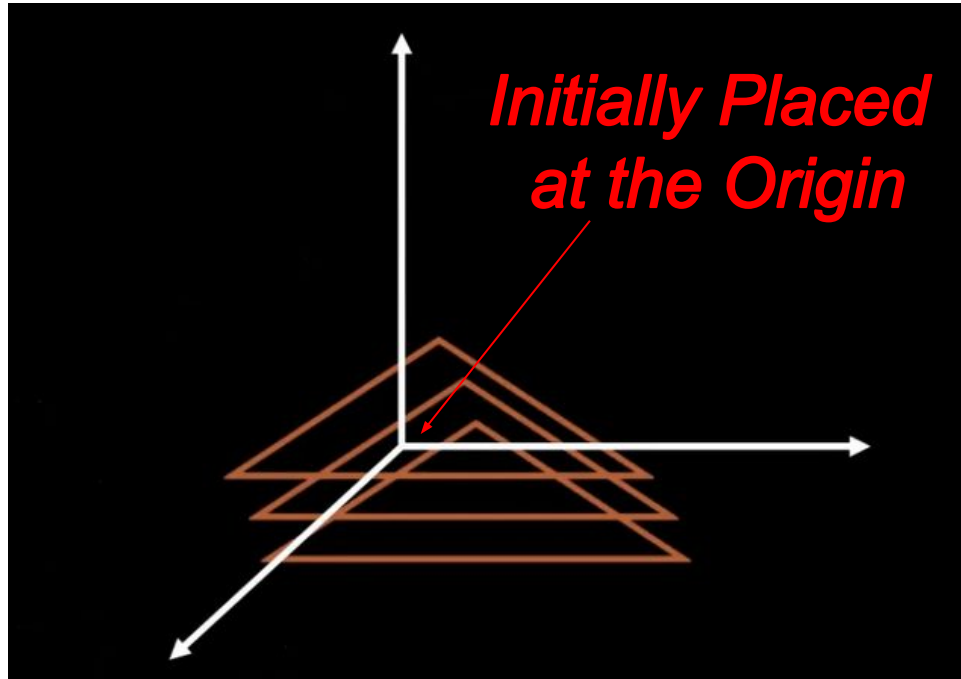
- Each residue is represented by a rigid **Local Frame** (red triangles).
- Structure prediction requires learning the **position and orientation** of these frames in **Euclidean Space**.

Structure Prediction Becomes Rotation and Translation of Local Frames



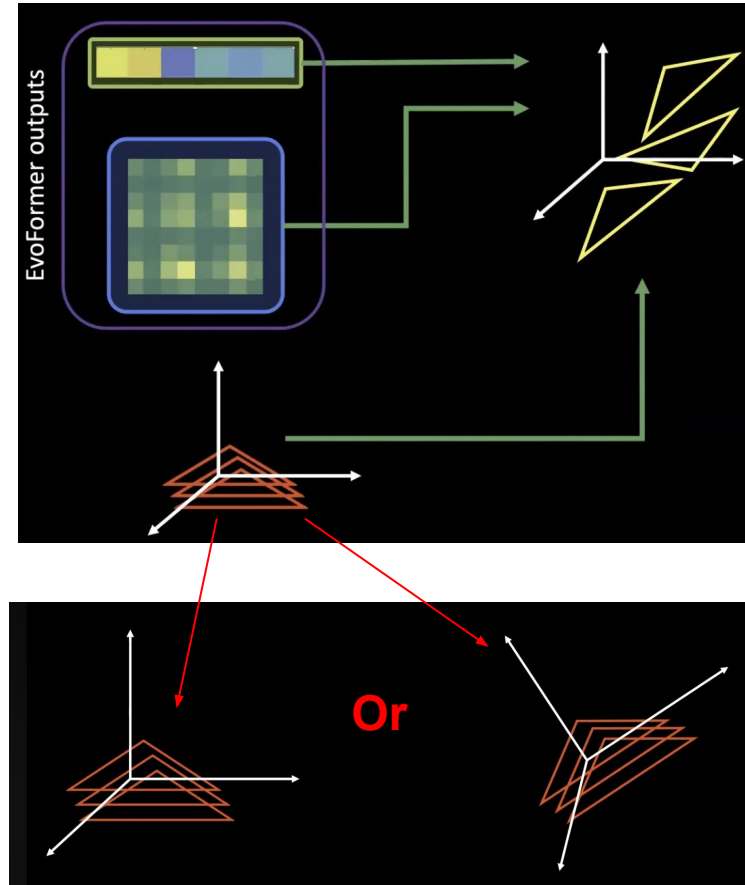
Our task is to learn the **rotation matrix R** and **translation vector t** for each of the triangles. There are 2 frames: **global coordinates frame** (white) and **local coordinates frame** (yellow).

“Black hole” Initial Conditions

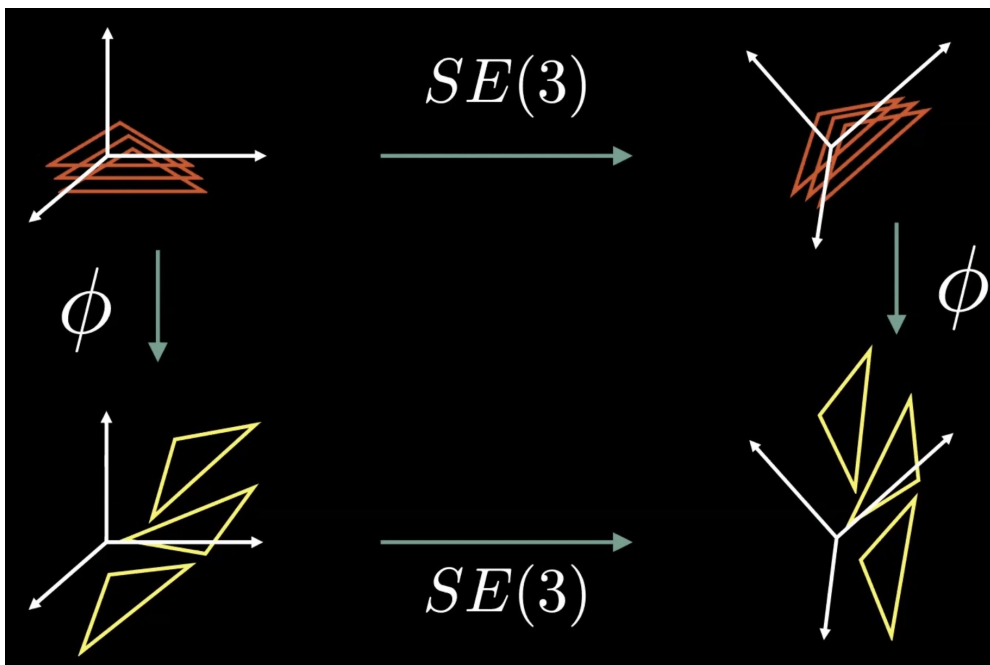


Initial conditions → **collapse all the triangles** (i.e. bag of words) **at the origin**, such that the $\mathbf{C}^a$ atom is sitting at the origin of the global reference frame, and all the local reference frames are oriented in the same direction as the global reference frame.

Problem Is: There Is No Meaningful Absolute Coordinate System



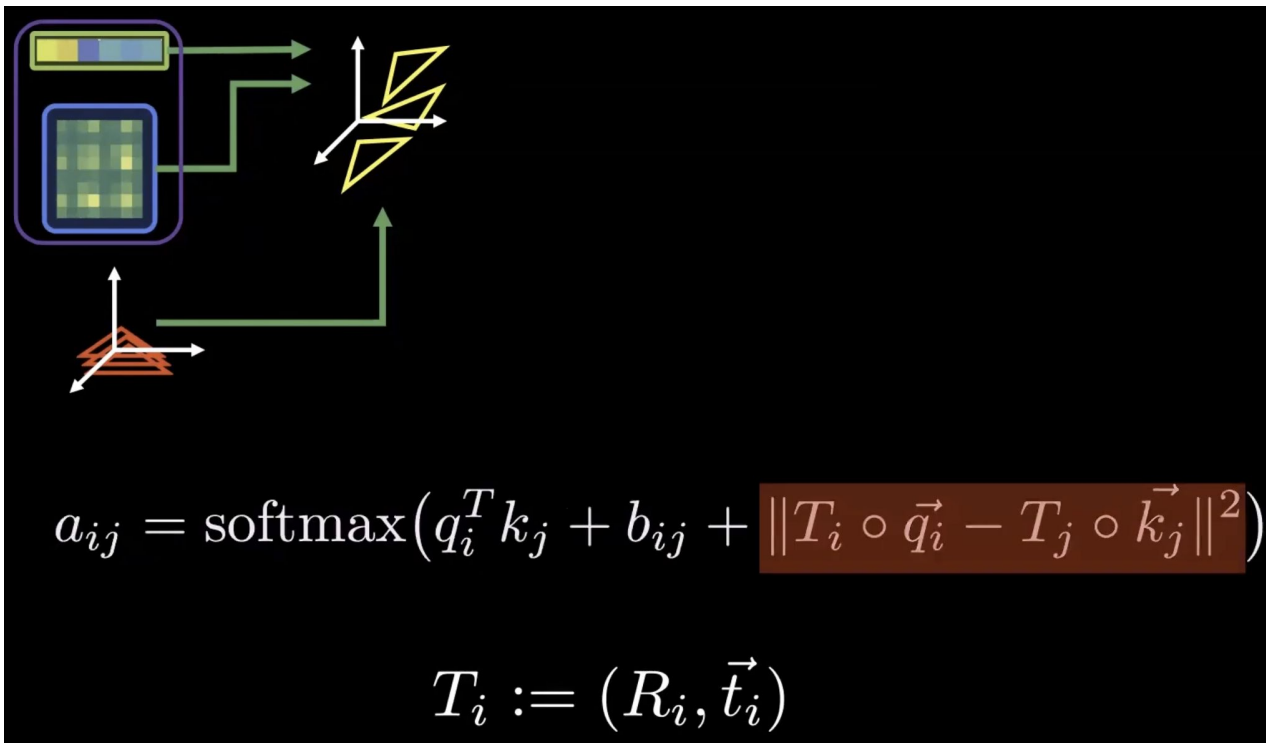
The Structure Module Must Be Rotation-Equivariant



The “**Evoformer**” operates on abstract feature representations with no geometric meaning. But the “**Structure Module**” must produce actual 3D coordinates. And since a protein has no preferred orientation in space, the structure module must be **SE(3)-equivariant**: *rotating the input must rotate the output by the same amount*. So both paths around the diagram must agree.

Invariant Point Attention Injects Geometry into Attention

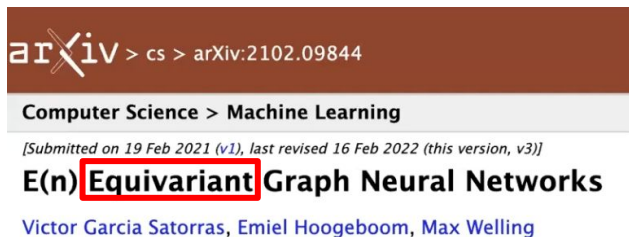
The geometrically aware attention formula becomes



This attention is called the “**Invariant Point Attention**”. It was novel at the time. We allow the queries and keys to live in the Euclidean space.

Invariant Point Attention Was Inspired by Equivariant Models

The type of equivariance we are using in AlphaFold 2 is **approximate**.

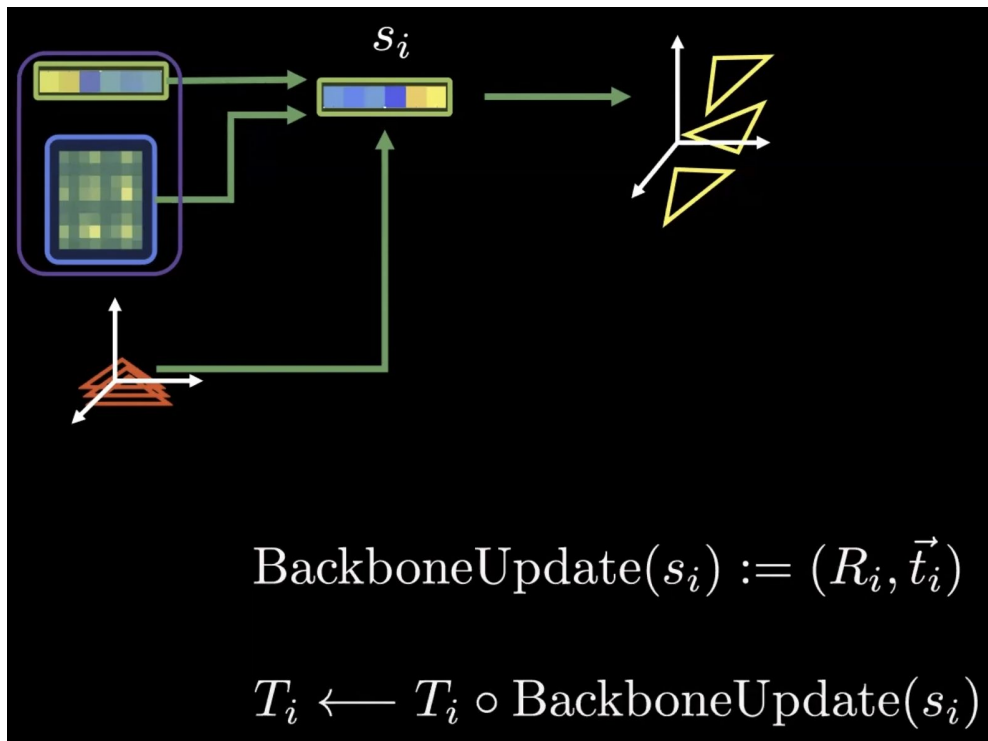


$$\mathbf{x}_i^{l+1} = \mathbf{x}_i^l + C \sum_{j \neq i} (\mathbf{x}_i^l - \mathbf{x}_j^l) \phi_x(\mathbf{m}_{ij})$$

$$T_i^{l+1} \leftarrow T_i^l \circ \text{BackboneUpdate}(s_i)$$

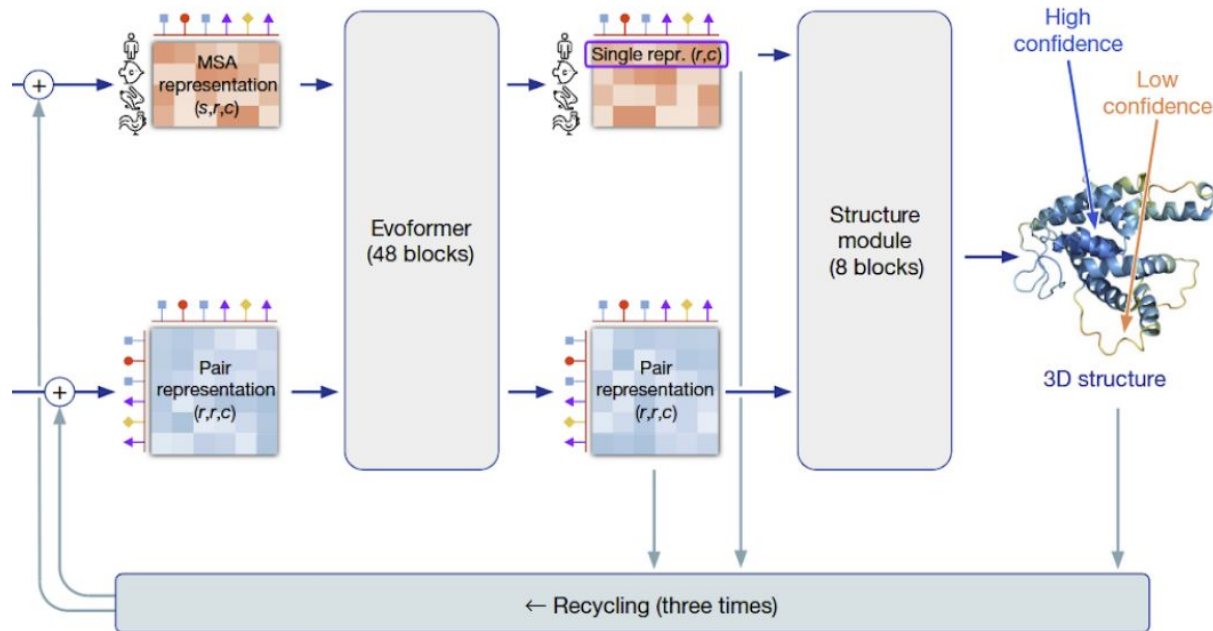
Note that we need not necessarily use sophisticated and mathematically principled techniques from **group theory** and **differential geometry**.

BackboneUpdate Predicts Local Frame Transformations



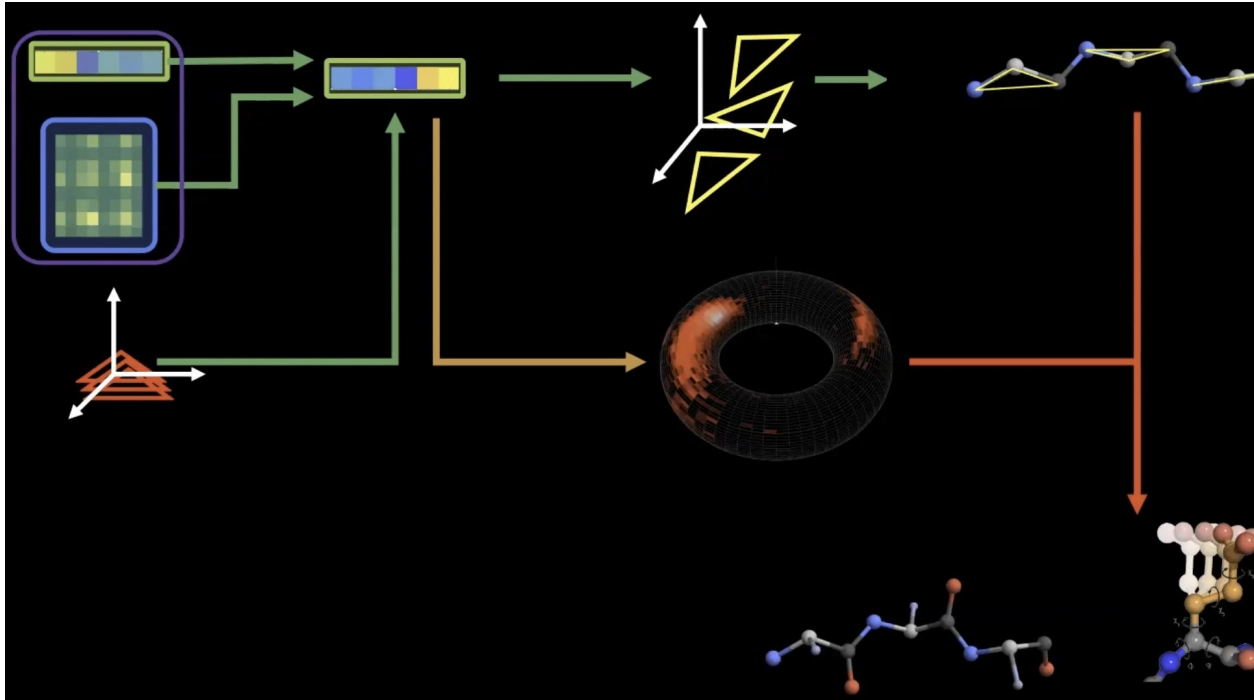
The updated single representation $\mathbf{S}_i$ is passed to **BackboneUpdate()**, which predicts the local **rotation** and **translation for each residue frame**.

Recycling Iteratively Refines the Predicted Structure



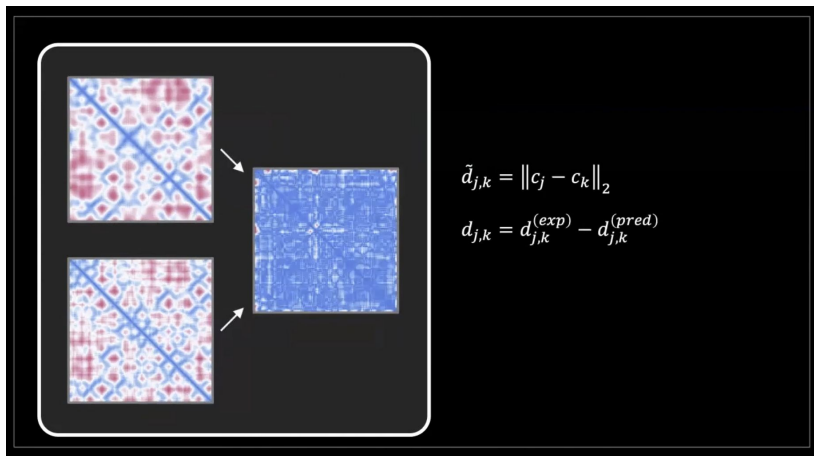
- There are **48 Evoformer blocks**
- Followed by **8 Structure module blocks**
- **Loss** is applied **after each block**
- **Recycling** $\approx$ iterative refinement (3 cycles)

Side-chains Are Predicted from the Same Structural Representation

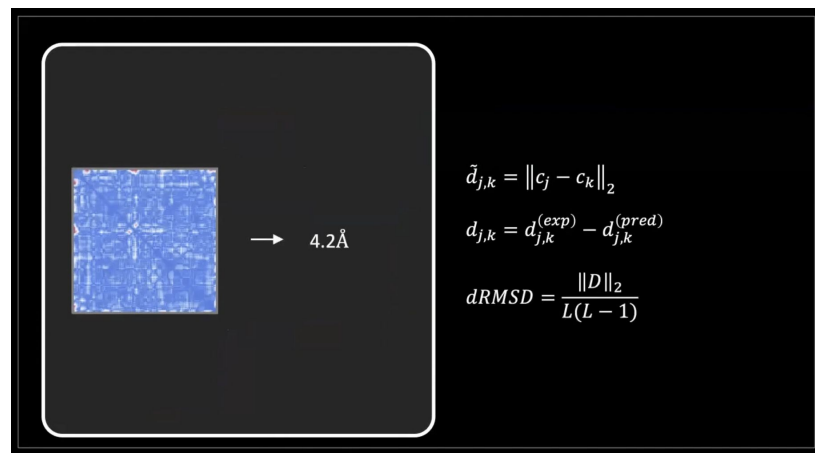


- The same single representation S_i is used to predict side-chain torsion angles.
- Because **side-chain geometry is largely rigid**, torsion prediction is sufficient.
- The system is trained **end-to-end**.

Structure Quality Can Be Measured Locally and Globally



Local Structure Accuracy (IDDT / LDDT)



dRMSD is a more global distance-based metric, but not the same as GDT (Global Distance Test)

- IDDT (i.e. Local Distance Difference Test) measures local structural accuracy.
- GDT (i.e. Global Distance Test) measures global structural similarity.

AlphaFold 2 Predicts Its Own Confidence via pLDDT

Model Confidence:

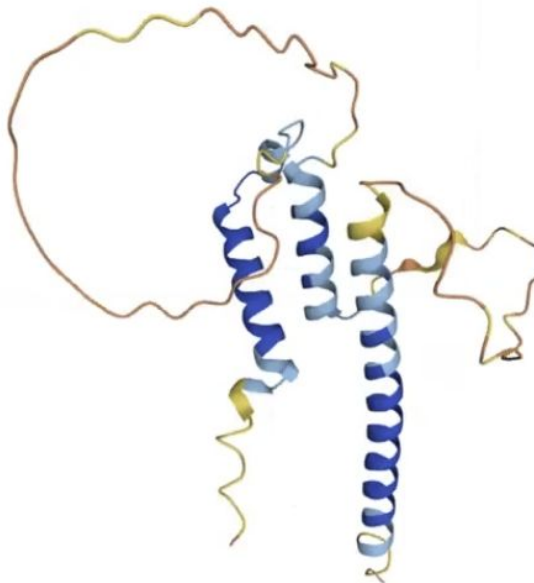
Very high (pLDDT > 90)

Confident (90 > pLDDT > 70)

Low (70 > pLDDT > 50)

Very low (pLDDT < 50)

AlphaFold produces a per-residue confidence score (pLDDT) between 0 and 100. Some regions with low pLDDT may be unstructured in isolation.



- **LDDT** = the **true (local)** evaluation metric
- **pLDDT** = the model's **predicted estimate** of that metric

The First Training Phase Combines Structural and Auxiliary Losses

$$\mathcal{L} = 0.5 \mathcal{L}_{\text{FAPE}} + 0.5 \mathcal{L}_{\text{aux}} + 0.3 \mathcal{L}_{\text{dist}} + 2.0 \mathcal{L}_{\text{MSA}} + 0.01 \mathcal{L}_{\text{conf}}$$

$\mathcal{L}_{\text{FAPE}}$ **FAPE loss** → geometry-aware loss invariant to global rigid motions

$\mathcal{L}_{\text{aux}}$ **Auxiliary loss** → per-block supervision during iterative refinement

$\mathcal{L}_{\text{dist}}$ **Distogram loss** → supervision on the pair representation

$\mathcal{L}_{\text{MSA}}$ **Masked MSA loss** → recovery of masked amino acids from evolutionary context

$\mathcal{L}_{\text{conf}}$ **Confidence loss** → supervision for confidence estimation

The Second Training Phase Adds Experimental and Violation Losses

$$\mathcal{L} = 0.5 \mathcal{L}_{\text{FAPE}} + 0.5 \mathcal{L}_{\text{aux}} + 0.3 \mathcal{L}_{\text{dist}} + 2.0 \mathcal{L}_{\text{MSA}} + 0.01 \mathcal{L}_{\text{conf}} + 0.01 \mathcal{L}_{\text{exp_resolved}} + 1.0 \mathcal{L}_{\text{viol}}$$

Experimentally resolved atom loss ($\mathcal{L}_{\text{exp_resolved}}$):

Not every atom in a protein is visible in experimental data. Some regions are floppy or disordered → they move around so much that X-ray crystallography or cryo-EM can't pin them down. **It would be unfair to penalize the model for getting these regions "wrong" when even nature hasn't settled on a fixed position for them.** So this loss only **counts errors where we actually have reliable experimental ground truth.**

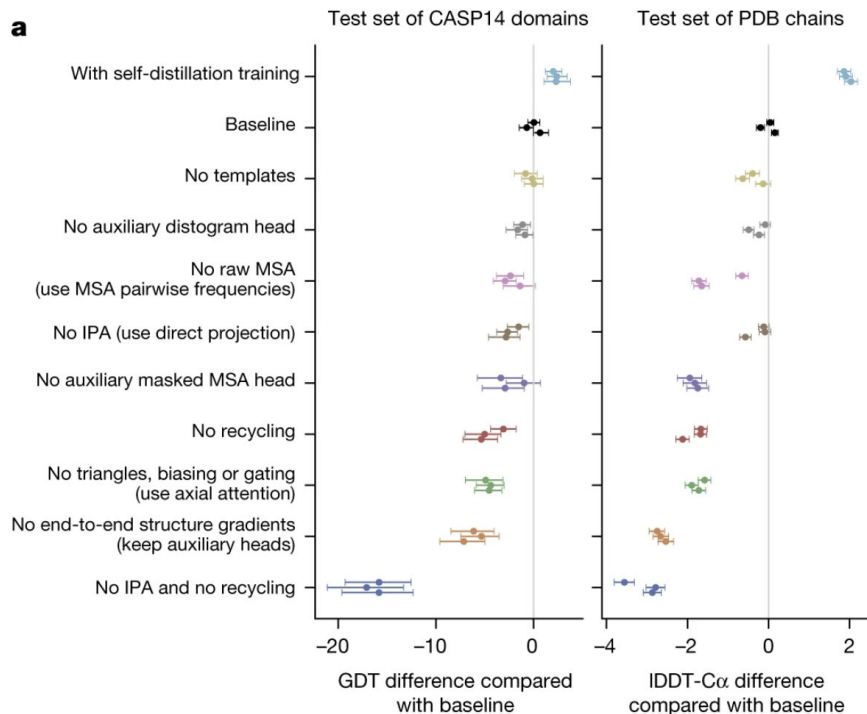
Violation loss ($\mathcal{L}_{\text{viol}}$):

This loss enforces basic physical and chemical reality. Even if the predicted structure looks good in terms of overall shape, it might contain local absurdities: **two atoms occupying the same space** (steric clashes), **bonds that are too long or too short** (invalid bond lengths), or **backbone angles that are physically impossible** (torsion outliers). This is like a spell-checker for chemistry → the model might write a grammatically correct sentence, but **this loss catches the typos at the atomic level. It has a high weight of 1.0 because these violations are non-negotiable:** a structure with atoms passing through each other is simply not a real protein, no matter how good the fold looks from a distance.

Ablation Studies Reveal Which Components Matter Most

Fig. 4: Interpreting the neural network.

From: [Highly accurate protein structure prediction with AlphaFold](#)



“Highly accurate protein structure prediction with AlphaFold” by J. Jumper et al., Nature, 2021

IPA and Recycling Are Critical for Performance

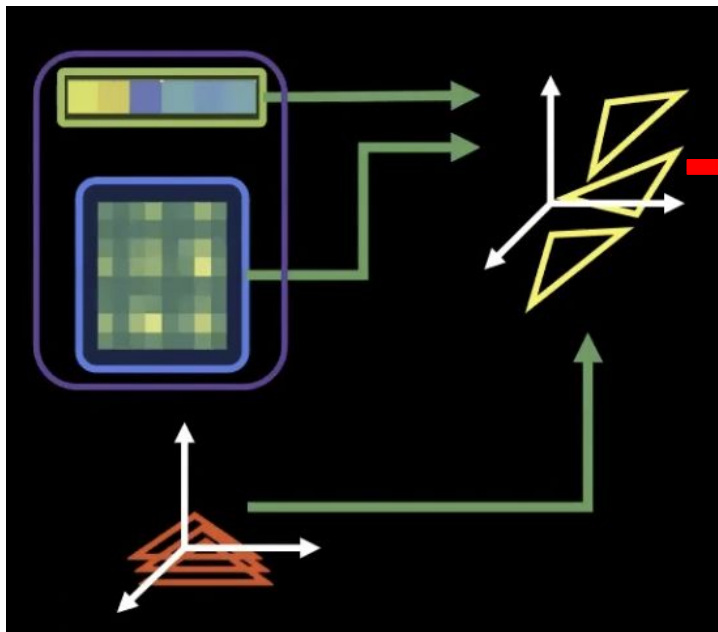
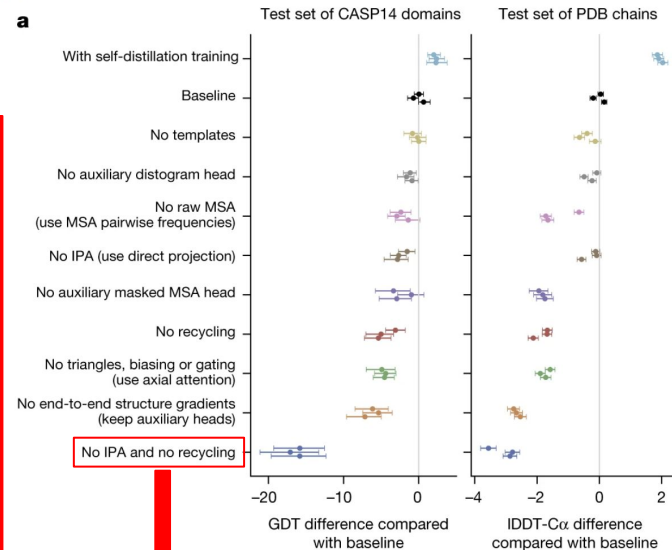


Fig. 4: Interpreting the neural network.

From: [Highly accurate protein structure prediction with AlphaFold](#)



$$a_{ij} = \text{softmax} \left(q_i^T k_j + b_{ij} \times \left\| T_i \circ \vec{q}_i - T_j \circ \vec{k}_j \right\|^2 \right)$$

Remember → In **AlphaFold 2**, **IPA** stands for **Invariant Point Attention**. It's one of the key architectural innovations that allows the model to reason directly in **3D space** while remaining consistent under rotations and translations.

OpenFold

OpenFold Prioritized Reproducibility and Trainability

Models	AlphaFold2	RoseTTAFold	OpenFold
Highly-accurate	✓	✓	✓
Platform	Jax	PyTorch	PyTorch
Data (MSAs)	×	×	✓
Trainable	×	×	✓

The screenshot shows the GitHub repository page for 'openfold' by 'aqlaboratory'. The repository is public and has 45 watches, 660 forks, and 3.3k stars. The 'About' section is highlighted with a red box, containing the text: 'Trainable, memory-efficient, and GPU-friendly PyTorch reproduction of AlphaFold 2'. Below the 'About' section, there are tags for 'protein-structure', 'pytorch', and 'alphafold2'. The repository also shows a merge pull request #560 from 'borisfom/fix-cueq-scaling' and a commit 'be2ec18' from 2 months ago with 1,612 commits.

Repository: **openfold** (Public)

Stats: Watch 45, Fork 660, Starred 3.3k

Branches: 7, Tags: 6

Search: Go to file

Buttons: Add file, Code

About: Trainable, memory-efficient, and GPU-friendly PyTorch reproduction of AlphaFold 2

Tags: protein-structure, pytorch, alphafold2

Recent Activity:

- jnwei Merge pull request #560 from borisfom/fix-cueq-scaling
- be2ec18 · 2 months ago
- 1,612 Commits

Files:

- .github Bump actions/checkout from 4 to 5 (6 months ago)
- docs Install docs added (3 months ago)

Ask DeepWiki for concrete Details

The screenshot shows the DeepWiki web application. The browser address bar displays the URL: `deepwiki.com/search/where-and-how-is-the-triangular-inequality-enforced-in-a-smooth-differentiable-way?mode=codemap`. The page title is "DeepWiki". The search results show a query for "aqlaboratory/openfold" with the question "Where and how is the triangular inequality enforced in a smooth differentiable way?".

The results are categorized into two main sections:

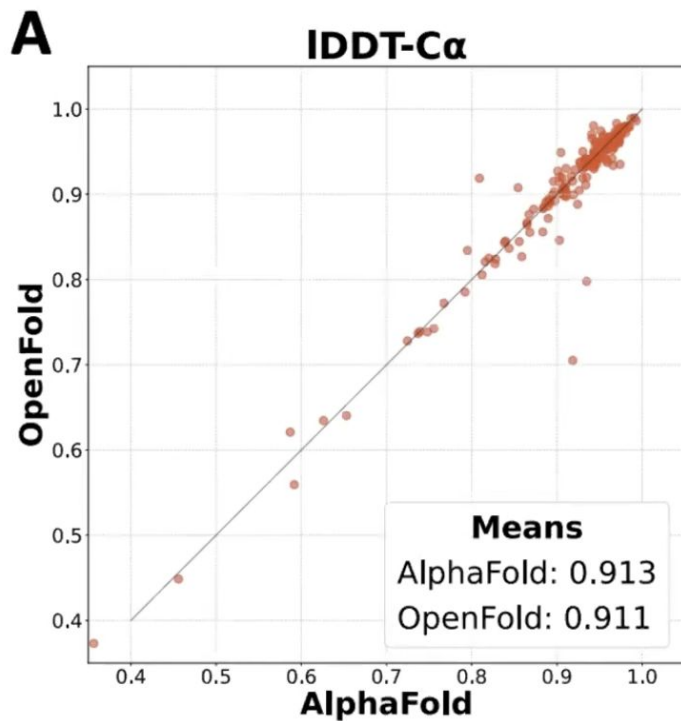
- Triangular Inequality Enforcement in OpenFold**
 - 1 Virtual Bond Computation Using Law of Cosines**
 - Core geometric constraint computation that translates bond angles into distance bounds. [See guide](#)
 - Parse bond lengths from file
 - Parse bond angles from file
 - Translate angles to virtual bonds
 - 1a Extract bond angle** (residue_constants.py:515)
`gamma = ba.angle_rad`
 - 1b Apply law of cosines** (residue_constants.py:516)
`length = np.sqrt(bond1.length**2 + bond2.length**2 - 2 * bond1.length * bond2.length * ...)`
 - 1c Store virtual bond** (residue_constants.py:538)
`residue_virtual_bonds(resname).append(Bond(ba.atom1_name, ba.atom3name, length, st...`
 - Return bonds, virtual_bonds, angles
 - 2 Distance Bounds Generation for Geometric Constraints**
 - Creating upper/lower bounds from virtual and standard bonds for enforcement. [See guide](#)
 - `make_atom14_dists_bounds()` function
 - Initialize bound arrays
 - 2a Van der Waals lower bounds**
`lower = atom1.radius + atom2.radius - overlap...`
 - Process all bond types
 - 2b Process all bond constraints**

- loss.py** (residue_constants.py)
- Copyright 2021 AlQuraishi Laboratory
- Copyright 2021 DeepMind Technologies Limited
- Licensed under the Apache License, Version 2.0 (the "License"); you may not use this file except in compliance with the License. You may obtain a copy of the License at <http://www.apache.org/licenses/LICENSE-2.0>
- Unless required by applicable law or agreed to in writing, software distributed under the License is distributed on an "AS IS" BASIS, WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied. See the License for the specific language governing permissions and limitations under the License.
- Imports: `import ml_collections, import numpy as np, import torch, import torch.nn as nn, from typing import Dict, Optional, Tuple`
- From openfold.np import `residue_constants`
- From openfold.utils.rigid_utils import `Rotation, Rigid`
- From openfold.utils.geometry.vector import `Vec3Array, euclidean_distance`
- From openfold.utils.all_atom_multimer import `get_rc_tensor`
- From openfold.utils.tensor_utils import `(tree_map, masked_mean, permute_final_dims)`
- Import `logging`
- From openfold.utils.tensor_utils import `tensor_tree_map`
- `logger = logging.getLogger(__name__)`
- Define `softmax_cross_entropy(logits, labels):`
 - `loss = -1 * torch.sum(labels * torch.nn.functional.log_softmax(logits, dim=-1), dim=-1)`
 - `return loss`
- Define `def simoid_cross_entropy(logits, labels):`

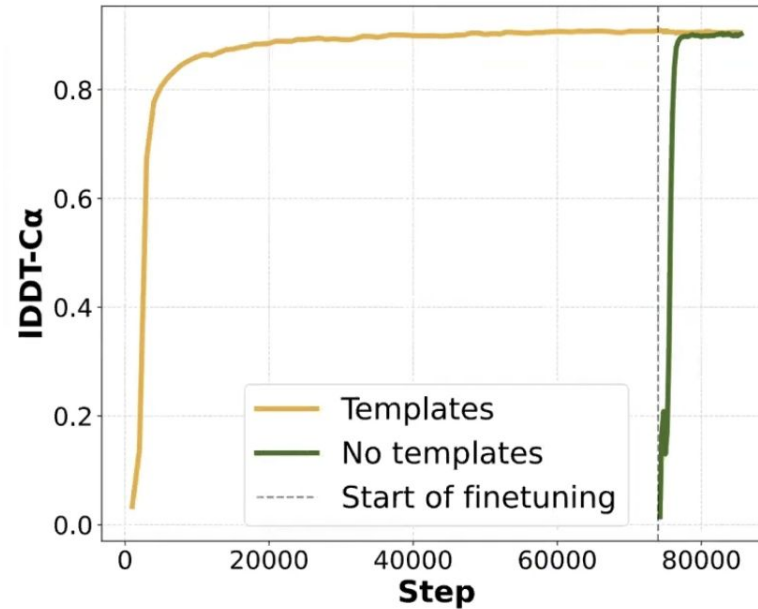
Append github username and repository name to the **deepwiki.com** URL → <https://deepwiki.com/aqlaboratory/openfold>

OpenFold Achieves Near-Parity with AlphaFold 2

CAMEO | Continuous Automated Model EvaluatiOn

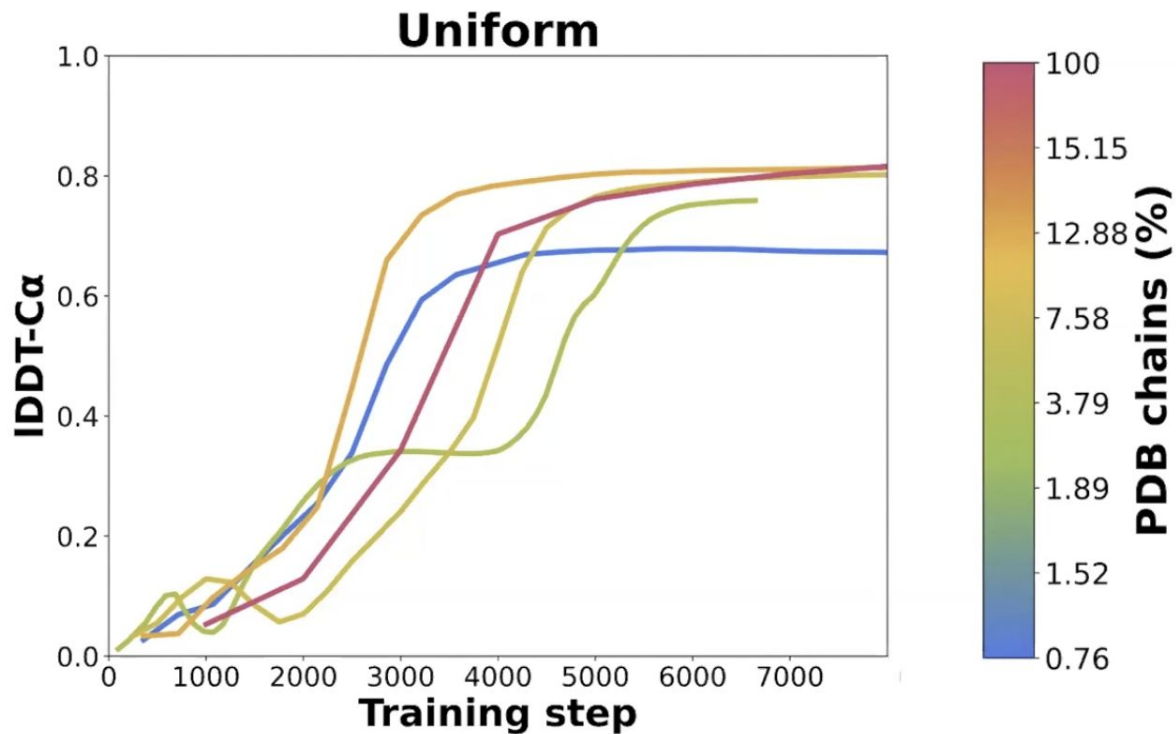


Most Performance Emerges Early in Training

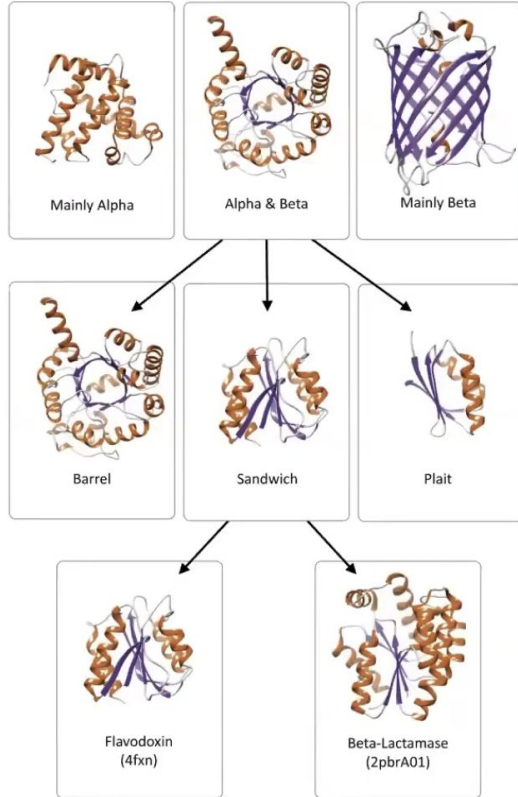


The model achieves **90% of the final metric** in only **3% of the training time**.

Performance with Deliberately Reduced Datasets



Protein Fold Space Is Rich but Still Constrained



Class

Architecture

Topology or Fold

***Nature is
parsimonious in
the structures it
allows***

Take-Home Messages

- AlphaFold 2 turned protein folding into a **learnable geometric reasoning** problem.
- Its success came from combining **co-evolution (1D)**, **pairwise reasoning (2D)**, and **triangle equivariant updates (3D)**.
- The architecture works because its inductive biases **match the biology and the geometry**.
- **Confidence prediction (pLDDT)** and reproducibility made the approach **scientifically useful**.

References

Theory material → <https://nazimbouatta.scholars.harvard.edu/lectures>

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Nobel Prize Talk → <https://www.youtube.com/live/HnT1VWzdFWc>

CASP challenge blog post → <https://deepmind.google/blog/alphafold-a-solution-to-a-50-year-old-grand-challenge-in-biology/>

OpenFold repository → <https://github.com/aqlaboratory/openfold>

“OpenFold: retraining AlphaFold2 yields new insights into its learning mechanisms and capacity for generalization”, by Ahdriz et al. → <https://www.nature.com/articles/s41592-024-02272-z>

DeepWiki → <https://deepwiki.com/aqlaboratory/openfold>

“MSA Transformer” by Rao et al. → <https://www.biorxiv.org/content/10.1101/2021.02.12.430858v1.full>

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“Do Transformers Really Perform Bad for Graph Representation?”, by Ying et al. → <https://arxiv.org/abs/2106.05234>

“CCNet: Criss-Cross Attention for Semantic Segmentation”, by Huang et al. → <https://arxiv.org/abs/1811.11721>

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Thank You!
Questions?